SENSOR AND INSTRUMENTATION (BOE 405)

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UNIT-1 SENSORS & TRANSDUCER

- Definition, Classification of transducers, Measurement of displacement using LVDT, Potentiometer & Optical Encoder; Measurement of force using strain gauge, Measurement of pressure using LVDT based diaphragm & piezoelectric sensor.

SENSORS & TRANSDUCER

A sensor is a device that detects and responds to some type of input from the physical environment. The input can be light, heat motion, moisture, pressure or any number of other environmental phenomena. The output is generally a signal that is converted to a human-readable display at the sensor location.

A transducer is a device that converts one form of energy into other, typically converting a physical quantity into an electrical signal

CLASSIFICATION OF SENSORS/ TRANSDUCERS

Sensors/ Transducers are classified into 7 major categories on the basis of their style of working:

Based on transduction principle: Primarily, the sensors/ transducers are classified on the basis of principle of working these are:

a. Resistive Type (RTD, Potentiometer, Strain, Gauge)
b. Inductive Type (LVDT, DC Motor)
c. Capacitive Type (Capacitive ear-phones)

Based on the quantity being measured:

a. **Temperature:** Resistant temperature detector RTD, Thermistor, thermocouple.
b. **Pressure:** Bourdon tube, diaphragm.
c. Force or Torque: Strain gauge.
d. Flow : Venturimeter (Petrol Pump)

CLASSIFICATION OF SENSORS/ TRANSDUCERS

Based on Power Requirement:

a. Active Sensor

b. Passive Sensor

Based on the power requirement, sensors can be classified as **active** and **passive** . **Active** sensor are those which do not require external power source for the functioning. They generate power within themselves to operate and hence called as self generating type. The energy for functioning is derived from the quantity being measured. For example piezoelectric crystal generate Electrical output when subjected to acceleration.

Passive sensors require external power source for their functioning. Most of the resistive inductive and capacitive sensors are passive, just as resistors, inductors and capacitors are called passive devices.

CLASSIFICATION OF SENSORS/ TRANSDUCERS

Based on Manufacturing Technology: It is also classified according to technology

a. It can be metallic
b. Optical
c. Semiconductor
d. Ultrasonic
e. Electromagnetic
f. Laser

Based on type of Output :

a. **Analog:** An analogue sensor converts the physical quantity being measured to the analog form. That is continuous in time. Thermocouple RTD strain gauge are called analogue sensors.
b. **Digital:** A digital sensor produces output in the form of Pulse. Encoders are example of digital sensor.

CLASSIFICATION OF SENSORS/ TRANSDUCERS

Based on stage of Application:

a. Primary (Direct contact with measurand)

b. Secondary

Based on Application:

a. **Industrial**

b. **Household**

PROCESS OF SELECTING A SENSORS

When selecting a sensor for a particular application, it is important to consider the following factors:

a. **Type of Sensing:** The parameter that is being sensed like temperature or pressure.

b. **Operating Principle:** The principle of operation of the Transducer/ Sensor.

c. **Power Consumption:** The power consumed by the transducer/ sensor will play an important role in defining the total power of the system.

d. **Accuracy:** The accuracy of the Transducer/ Sensor is a key factor in selection.

e. **Resolution and Range:** The smallest value that can be sensed and the limit of measurement are important.

f. **Cost:** Depending on the cost of application, a low cost Transducer/ Sensor or high cost sensor can be used.

CHARACTERISTIC OF SENSORS

Static Characteristic

- Accuracy
- Precision
- Resolution
- Range
- Sensitivity
- Linearity etc.

Dynamic Characteristic

- Speed of response
- Dynamic error

a. **Static Characteristic** The set of criteria defined for the instruments which are used to measure the quantities which (do not varying with time) are slowly varying with time or mostly constant

b. **Dynamic Characteristic:** The set of criteria defined for the instruments which are changes rapidly with time.

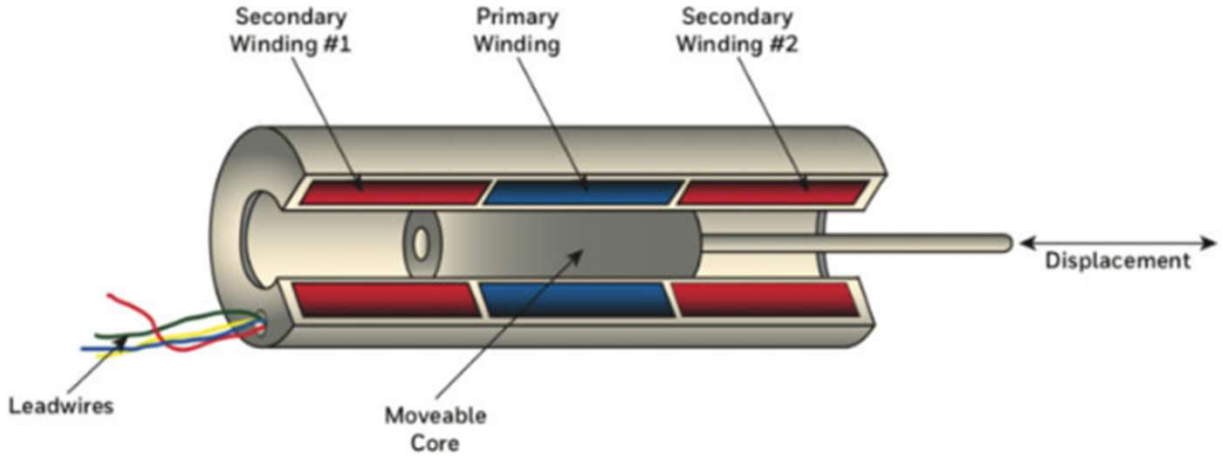
LVDT

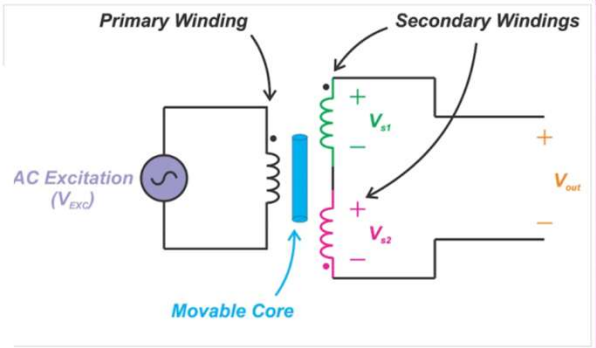
Linear Variable Differential Transformer is a common type of electromechanical transducer that can covert the rectilinear motion of an object to which it is coupled mechanically into a corresponding electrical signal

Construction: LVDT is a transformer consisting of one primary winding P and two secondary winding S1 & S2 mounted on a cylindrical former.

A movable soft iron core is placed inside the former. Actually the movable core made of nickel – iron.

Both the secondary winding are connected in series But in phase opposition.





An LVDT contains a primary winding and two secondary windings wound around a hollow form. A ferromagnetic core sits inside the bore and travels back and forth along the windings.

Image credit: Honeywell International

WORKING OF LVDT

Since the primary winding of Linear Variable Differential Transformer (LVDT) is supplied with AC supply, it produces an alternating magnetic flux in the core which in turn link with the secondary winding S1 and S2 to produce emf due to transformer action. The emf produced in secondary winding S1 and Es1 and that in S2 is Es2. The magnitude of Es1 and Es2 will depend upon the magnitude of rate of change of flux **(dϕ/dt) acc to faradays law.**

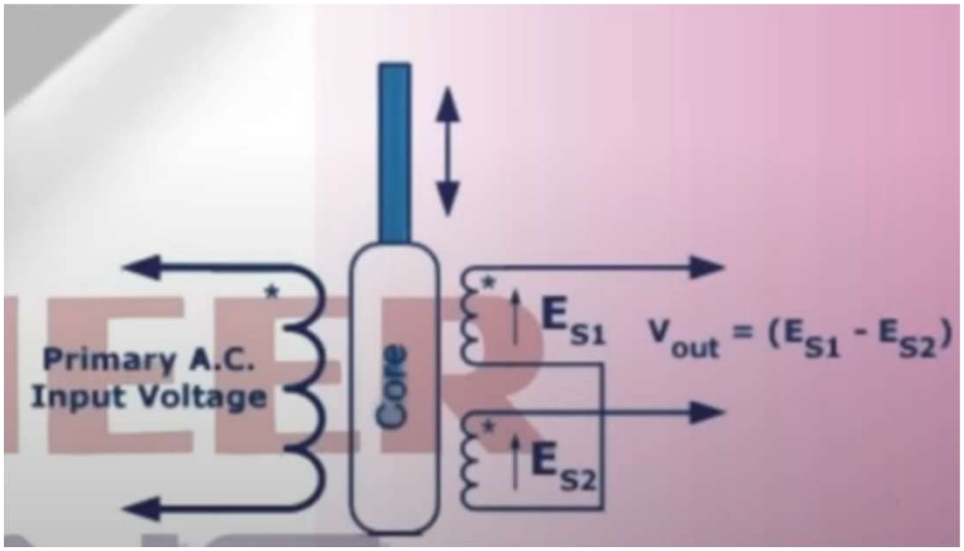
Thus faster the movement of core the greater will be the magnitude of emf induced in secondary windings.

Net output voltage E0 of the LVDT is: E0 = Es1 –Es2

Core at center: Es1 = Es2 and hence net ouput voltage E0 of LVDT = 0

Core at left Es1 > Es2 and net output voltage E0 = (Es1 – Es2) = Positive.

Core at right Es2 >Es1 and hence net output voltage E0 = (Es1 – Es2) = Negative.



ADVANTAGES AND DISADVANTAGES

Advantages:

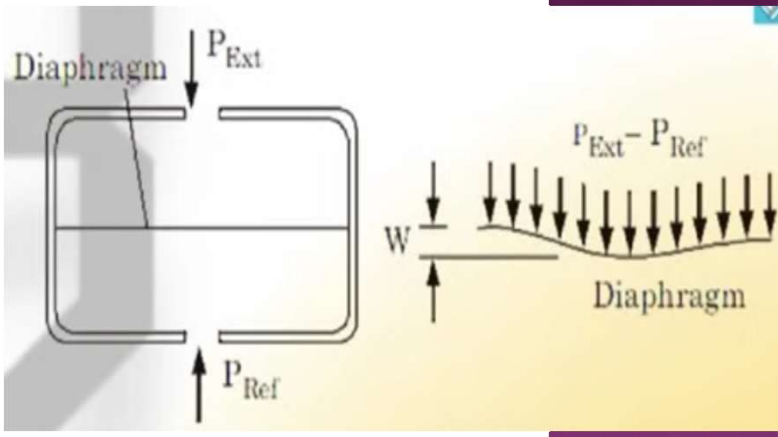
1. High output and high sensitivity: The LVDT gives a high output and a high sensitivity.
2. Ruggedness: These transducers can usually tolerate high degree of shock and vibrations especially when the core is spring loaded without any adverse effects.
3. Low hysteresis
4. Low power consumption: Most of LVDTs consumes very less power

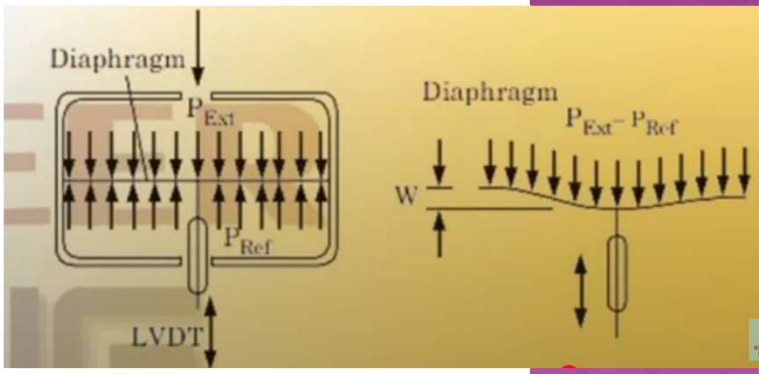
Disadvantages:

1. Very high displacement is required for generating high voltages.
2. Shielding is required since it is sensitive to magnetic field.
3. The performance of the transducer gets affected by vibrations.
4. It is greatly affected by temperature changes.

MEASUREMENT OF PRESSURE USING LVDT

- The diaphragm pressure gauge uses the elastic deformation of diaphragm (i.e. membrane) instead of a liquid level to measure the difference between the unknown pressure and a reference pressure.
- A typical diaphragm pressure gauge contains a capsule divided by a diaphragm
- One side of diaphragm is open to the external targeted pressure, $P_{Ext.}$, and other side is connected to known pressure, P_{Ref}. the pressure difference $P_{Ext} - P_{Ref}$, mechanically deflects the diaphragm.
- The membrane of the diaphragm is connected to LVDT. Due to application of pressure on diaphragm, a displacement is applied on core or primary winding of LVDT. As a result a voltage is induced in the secondary winding of LVDT. This induced voltage is directly proportional to applied pressure.





STRAIN GAUGE

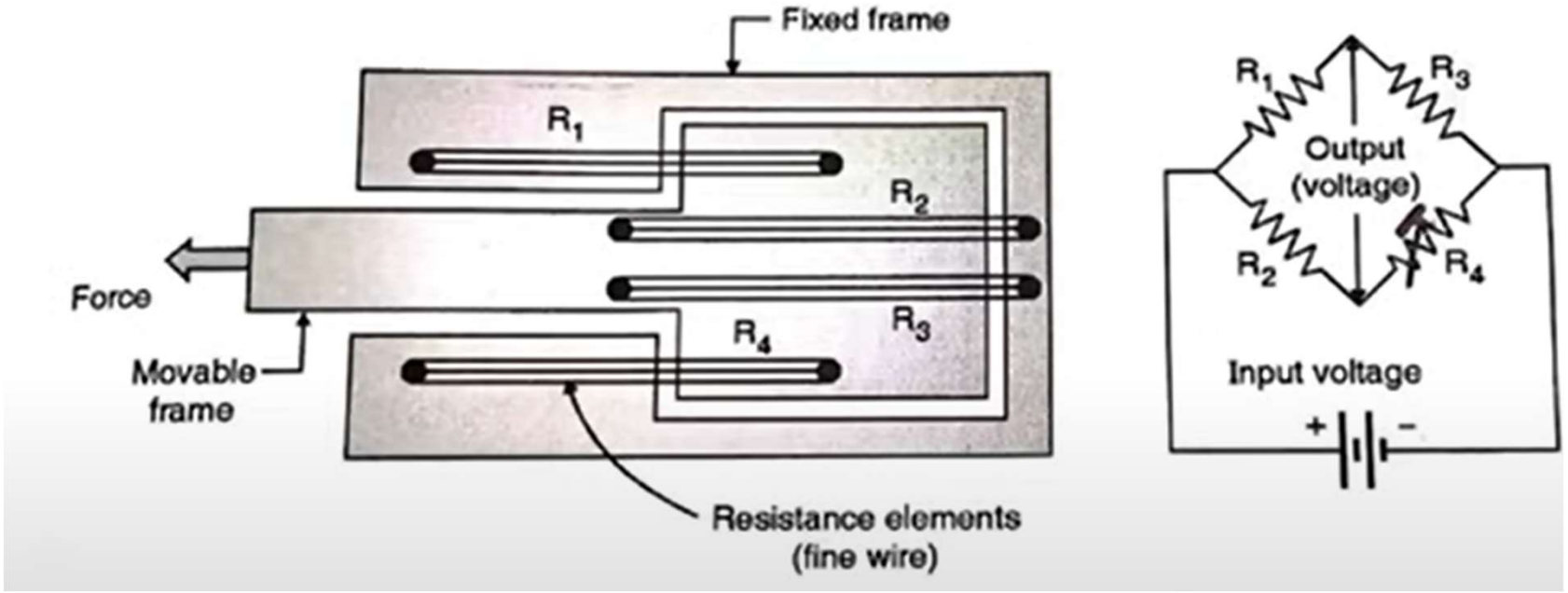
- Device to measure these small changes in dimensions are called strain gauges. A strain gauge, a device whose electrical resistance varies in proportion to the amount of strain in the device.
- A strain gauge works on the principle of electrical conductance and its dependence on the conductor's geometry. Whenever a conductor is stretched within the limits of its elasticity, it doesn't break but, gets narrower and longer.
- Strain gauge is a passive transducer that converts a mechanical elongation or displacement produced due to a force into its corresponding change in resistance. It uses the variation in electrical resistance in wires to sense the strain produced by a force on the wire.

MEASUREMENT OF FORCE USING STRAIN GAUGE

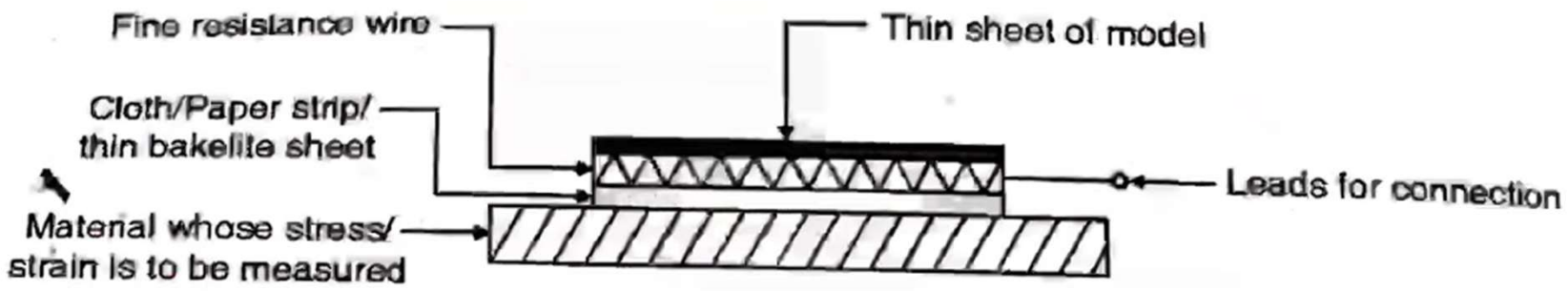
Following are the major types of strain gauge:

- Unbonded strain gauges
- Bonded strain gauges

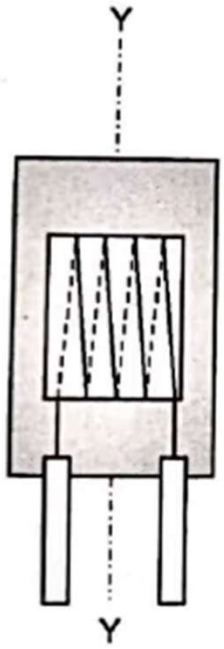
UNBONDED STRAIN GAUGES



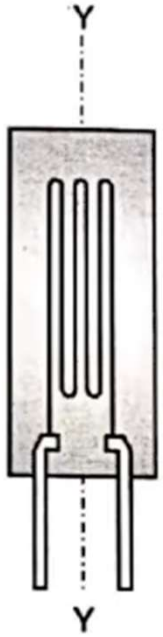
BONDED STRAIN GAUGES



DIFFERENT CONSTRUCTION OF BONDED STRAIN GAUGE



(i) Wamp around wire strain gauge



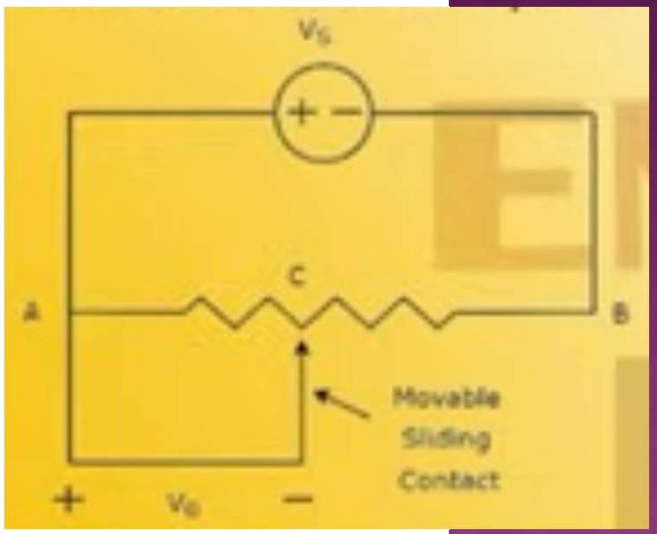
(ii) Flat-gird strain gauge



(iii) Foil strain gauge

MEASUREMENT OF DISPLACEMENT USING POTIENTIOMETER

The potentiometer is an electrical type of transducer or sensor and it is of resistive type because it works on the principle of wire changes.

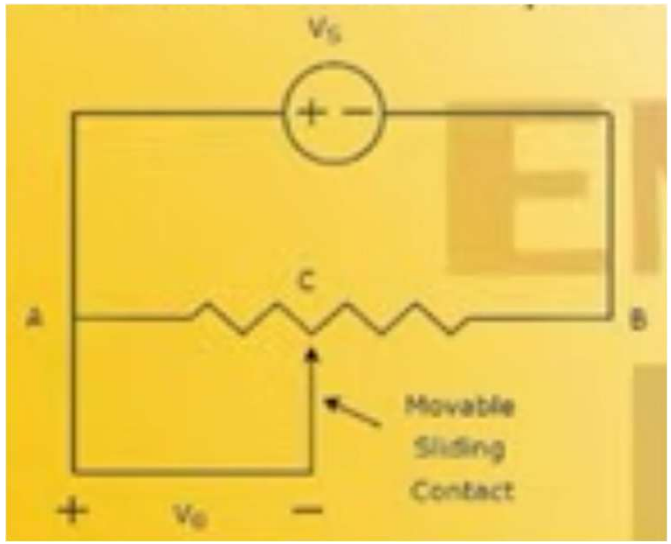


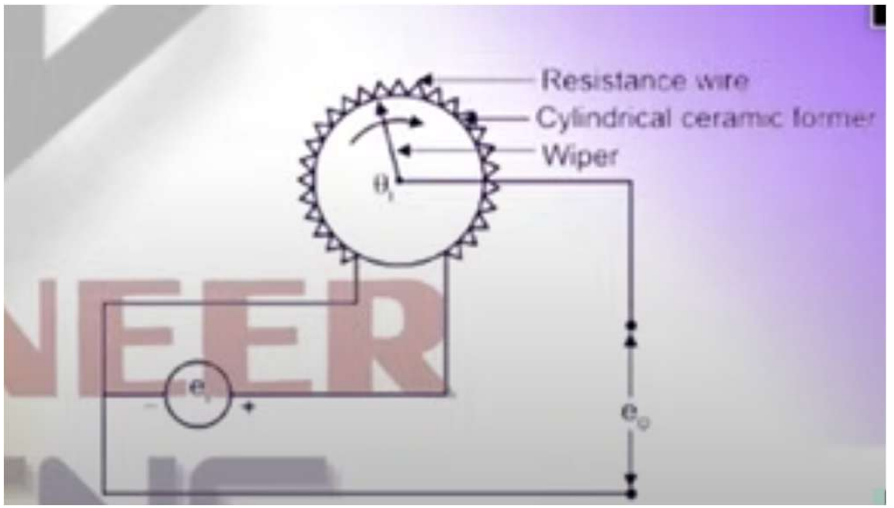
Construction

The above circuit consists of a potentiometer and voltage source, V_S we can say that these two are connected in parallel with respect to the points A & B. Potentiometer has a sliding contact, which can be varied. So the point C is a variable one. In above circuit the output voltage, V_0 is measured across the points A & C.

WORKING

- Therefore, we should connect the body whose displacement is to be measured to the sliding contact. So, whenever the body moves in a straight line, the point C also varies. Due to this, the output voltage, V_0 also changes accordingly.
- We can find the displacement by measuring the output voltage V_0.

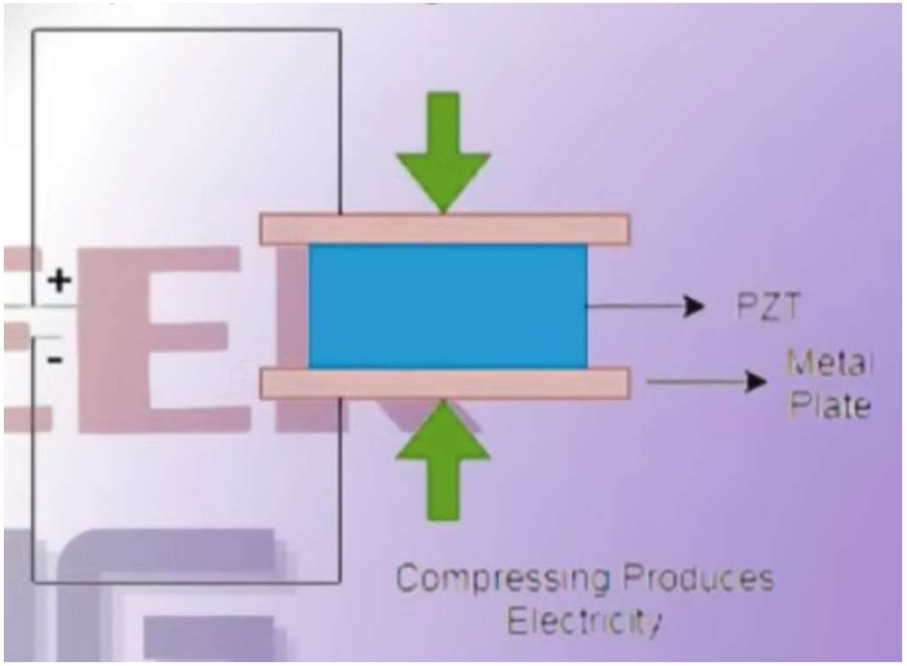




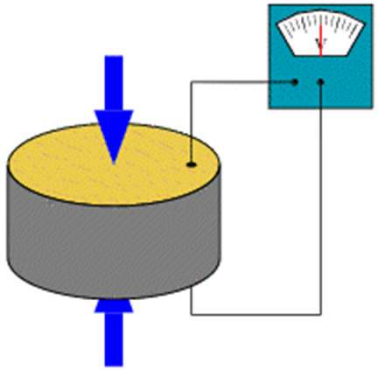
PIEZOELECTRIC SENSOR

A sensor that utilizes the piezoelectric effect, to measure changes in acceleration, strain, pressure, and force by converting them into electrical charges is called as a piezoelectric sensor. Piezo is a Greek word which means 'press' or 'squeeze'.

Piezoelectricity is phenomena where electricity is generated if the mechanical stress is applied to a material. This piezo electricity produced is proportional to the stress given to substrates of the strong piezoelectric crystal.



PIEZOELECTRIC EFFECT



Initially, in piezoelectric crystal, the charges are exactly balanced and the effect of charges cancel out each other and hence no net charge will be found on the crystal surface.

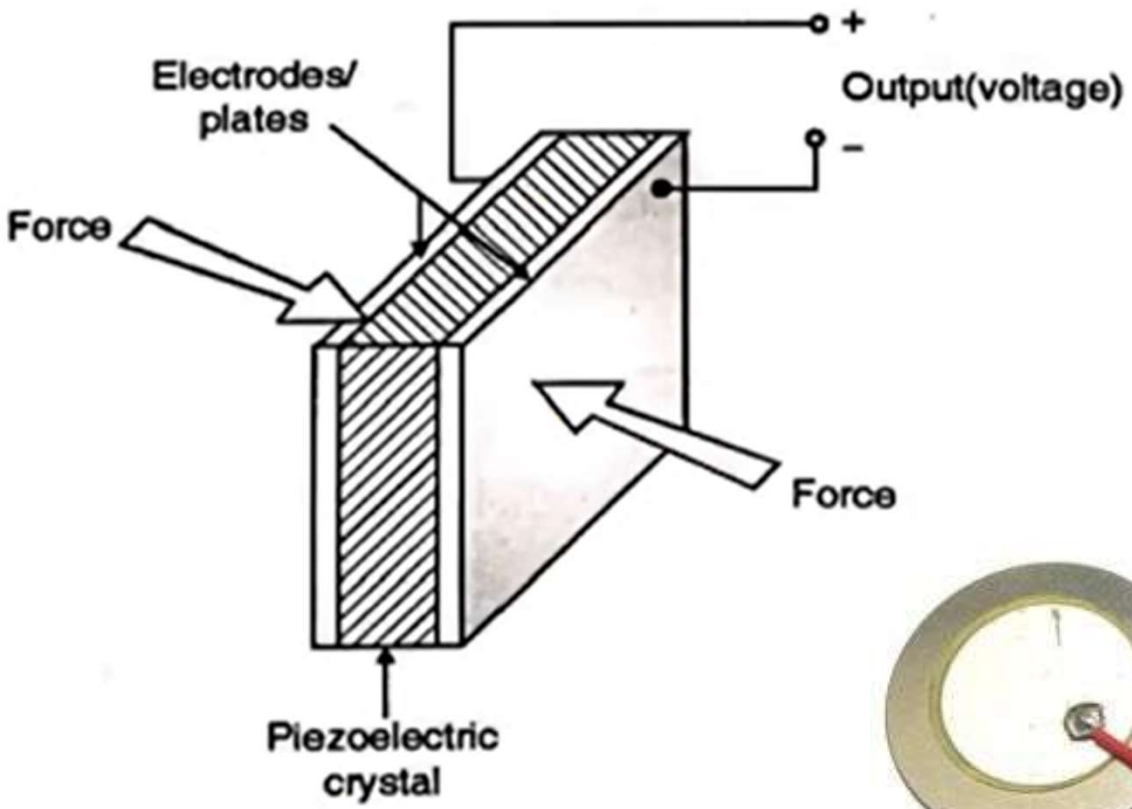
When the crystal is squeezed the charges in crystal become unbalanced which makes net Positive and negative charge to appear on the opposite faces of the crystal. Therefore, by squeezing the crystal, voltage is produced across the opposite face, and this is known as piezoelectricity. To make it a complete circuit, two phases are connected together and current is passed through.

For Example, Lighters. Gas burners lighter utilizes piezoelectric effect. It produces electric pulse due to the force developed by sudden impact of trigger over the inside.

PIEZOELECTRIC MATERIALS

- Quartz is used commonly. However its output is small, so amplification is needed.
- Rochelle salt provides highest output. But it has to be protected from moisture.
- Barium titanate material can be formed into variety of shapes and Natural crystal have higher mechanical and thermal stability.

PIEZOELECTRIC SENSOR



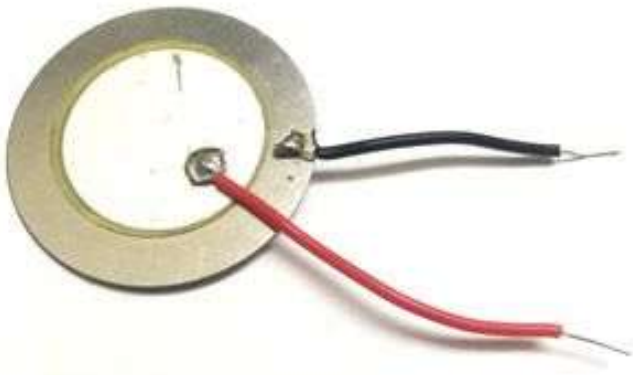


Figure: Piezoelectricity Crystal

APPLICATIONS

- It is also used in medical diagnostics.
- It is used in electric lighter used in kitchens. The pressure made on piezoelectric sensor creates an electric signal which ultimately causes the flash to fire up.
- They are used for studying high-speed shock waves and blast waves.
- It is used infertility treatment.
- It is used in inject printers.

Advantages

1. No need for an external emf.
2. Easy to handle and use as it has small dimensions.
3. High-frequency response it means the parameters change very rapidly.

Disadvantages:

1. It is not suitable for measurement in static condition.
2. It is affected by temperatures.
3. The output is low so some external circuit is attached to it.
4. It is very difficult to give the desired shape to this material and also desired strength.

OPTICAL ENCODER

The optical encoder is a transducer commonly used for measuring rotational motion. It consists of a shaft connected to a circular disc, containing one or more tracks of alternating transparent and opaque areas. A light source and an optical sensor are mounted on opposite sides of each track. As the saft rotates, the light sensor emits a series of pulses as the light source is interrupted by the pattern on the disc. This output signal can be directly compatible with digital circuitry.

Both displacement and velocity of the moving disk may be sensed in this manner.

